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RESEARCH ARTICLE

Indoor Localization Using Smartphone Magnetic Sensor Data: A Bi-LSTM Neural Network Approach

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ABSTRACT Nowadays, smartphone-based indoor localization has gained significant attention across various research fields, industrial sectors, and practical applications. However, accurate indoor localization using smartphones is a challenging issue due to the presence of noise and crowded localization environments. To overcome these problems, we introduce a new system for localization in indoor locations using real magnetic data collected from a smartphone. In this system, we first normalize the dataset. After that, the normalized magnetic data are filtered by using the Empirical Mode Decomposition (EMD) method. Finally, we present a Bidirectional Long Short-Term Memory (Bi-LSTM) neural network for indoor localization using fingerprinting technique. We use three datasets in three different locations in the Faculty of Engineering. Results show that Bi-LSTM neural network has achieved the best localization performance among Feed Forward Neural (FFN) Network, Gated Recurrent Unit (GRU) neural network and LSTM neural network. Compared to the performance of the mentioned neural networks, the proposed system had the least mean and standard deviation (SD) root mean square error (RMSE) for indoor localization in the datasets. According to the results obtained with the proposed system, mean and SD RMSE for the network and embedded systems research laboratory location on the second floor of the Engineering Faculty is 0.1244 m and 0.1070 m, for the corridor location on the same floor is 0.0251 m and 0.0166 m, and for first floor location of this faculty is 0.0192 m and 0.0165 m, respectively. Therefore, the proposed method can be used practically in the real world.

INDEX TERMS Bi-LSTM neural network, indoor localization, magnetic sensor, smartphone.

I. INTRODUCTION

In the present era, The Internet of Things (IoT) has made many of our daily applications smart and is used in a variety of smart applications such as smart transportation, smart agriculture, smart health, and smart home and because of its various uses, it is becoming very popular in all the developed countries of the world [1], [2], [3]. Due to the wide applications of smart services in the location and the significant increase in the demand for the people positioning, we have studied about localization. Localization is divided into two groups: indoor and external localization. But due to the various applications of indoor localization, including tracking elderly people [4] or patients who suffer from Alzheimer's disease [5], indoor localization has become an important part of people's daily lives [6], [7], [8]. Determining the location inside the building is divided into two categories based on the infrastructure and independent of the infrastructure [9], [10], [11], [12], [13], [14], [15]. In infrastructure-based methods require infrastructure such as the Internet, which includes Infrared [9], WLAN [10], Ultrasonic [11], Bluetooth [12], and Radio Frequency Identification (RFID) [13] methods. However, these methods are not feasible for localization in indoor environments, because it is difficult and expensive to deploy the infrastructure for them. Furthermore, these localization methods do not use smartphones. Infrastructure

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independent methods such as lasers [14] and vision [15] do not require additional infrastructure. one of the location determination methods independent of the infrastructure inside the building, which has been widely noticed by researchers, is the location determination method using smartphone sensors because determines the location continuously. This method is very suitable in some special situations such as fire or power outage when infrastructures such as Wireless Fidelity (Wi-Fi) are not able to serve. Indoor localization based on smart phones is becoming important in various applications such as locating people in public places including: nursing homes, airports and stores [16]. indoor localization with smartphones uses sensors embedded in them [17].

Smartphone datasets commonly include data from accelerometers, gyroscopes, and magnetometers [18]. These sensors have the advantages of small size, light weight and low energy consumption. Magnetometers are particularly useful for indoor localization because they are infrastructure-independent and require no specialized hardware [19], [20], [21]. According to existing research, the way the smartphone is placed has a great influence on the magnetic field and can be used in common modes including: ''navigating'', ''listening to the call'' and ''phone swinging''. Navigation is the mode where the phone is placed in the hand in front of the user's body. The user moves in different directions but does not change the orientation of the smartphone. In listening mode, the smartphone is placed on one side of the face and the camera is on top of the smartphone. The smartphone screen is facing the user. In phone swing mode, the phone swings in the user's hand while walking [22]. This paper focuses on data collected with the smartphone in navigating mode.

Indoor localization research faces challenges in achieving desired accuracy due to input data and smartphone maintenance. The signals that have been used for indoor localization so far include: Received Signal Strength (RSS) [6], [12], Bluetooth Low Energy (BLE) [23], smartphone signals [16], [17], [18] and etc. To the best of our knowledge, there is a limited number of works performed on indoor localization using smartphone magnetic signal. For example, indoor localization prediction using magnetic signal by LSTM networks [21], deep LSTM [24] and A multi-scale Temporal Convolutional Network (TCN) and LSTM network [25] is intended for only one floor. In another study, indoor localization has been investigated by combining Inertial Measurement Unit (IMU) data including magnetic data from smartphone sensors and Wi-Fi data using K Nearest Neighbor (KNN) method in several floor [22], but the accuracy of indoor localization is very poor.

Selecting an appropriate location prediction model can play a key role in designing indoor location predictors. Since neural networks have a high capability in network training and location prediction, they are used for indoor magnetic localization. Given that neural networks can train the network automatically and can also predict the target in real time [8], [9]. Therefore, we use neural networks to predict indoor location. Neural networks are very well known and widely used and have been successfully implemented in different practical applications such as spatiotemporal $PM_{2.5}$ forecasts based on atmospheric chemical transport model output and monitoring data [26], earthquake location prediction [27], noise reduction and classification of v-band receiver signals [28], Prediction of hysteresis response of steel braces [29], Groundwater level forecasting [30], Predicting the ship's path [31], and Predicting concrete corrosion in wastewater treatment plant inlet chambers [32]. Among the advantages of the Bi-LSTM neural network compared to other neural networks is the bidirectional prediction of the Bi-LSTM neural network, which processes and predicts data in both forward and backward directions, and has more accurate long-term prediction and less error compared to traditional RNN networks. In addition, LSTMs are designed to reduce the vanishing gradient problem common in standard RNNs, allowing Bi-LSTMs to learn and predict without losing important information from previous time steps. Also, in tasks where the relationships between elements are complex and require understanding from multiple time steps, the Bi-LSTM architecture allows for deeper analysis and better feature extraction. Therefore, we use from the Bi-LSTM neural network for indoor localization because is a very powerful neural network. This Neural Network has been successfully used in a variety of practical applications such as day-ahead peak load forecasting [33], Taxi-Passenger's Destination Prediction [34], Intelligent Vessel Trajectory Prediction [35], Short-Term Power Load Forecasting [36] and Epilepsy EEG Prediction [37]. Therefore, according to the capability of this neural network, we use it to estimate the position of persons.

The contribution done in this research is summarized as follows:

A. COLLECTION OF UNIQUE DATASETS OF THE FACULTY OF ENGINEERING

Considering that for collecting indoor positioning data, the type of input data and its related costs must be considered. In this study, we use magnetometer data of smartphone sensors due to their availability and portability. In addition, we do not pay any fee for their collection. The data of this study include location data and magnetometer data collected from the magnetometer sensor of Samsung Galaxy A3 smartphone in the Faculty of Engineering of this university, which include: 254, 462 and 685 data respectively from a small location in the Network and Embedded Systems Research Laboratory with an area of 8.2×5.7 square meter on the second floor of this faculty with several spiral paths, a large closed location in the corridor of the same floor which is a long and empty corridor, and a large location on the first floor of this faculty.

B. DATA PREPROCESSING

For data preprocessing, we first normalize the data to avoid overfitting. Then, we filter the normalized magnetic data

using the EMD algorithm to remove noise in the magnetic data.

C. BI-LSTM NEURAL NETWORK TRAINING USING THE LOCATION DATA OF THIS RESEARCH AND MAGNETOMETER DATA

we used most of the processed magnetic data and the corresponding position data for training and less than 50 data in each of the three locations for testing so that the Bi-LSTM neural network can be better trained and make better prediction. Then, we used the Bi-LSTM neural network to train the processed magnetic data and position data.

D. INDOOR LOCATION PREDICTION

We use the trained network and smartphone magnetic test data to predict the location of people indoors. Then, we use the mean and SD RMSE between the actual location and the predicted location to evaluate the accuracy of the proposed method.

This paper is organized as follows. Section II presents related works. Section III describes the structure of the proposed system, namely indoor localization using Bi-LSTM neural network. Section IV compares and evaluates the results of Bi-LSTM neural network with other neural networks in this paper. Finally, Section V discusses the conclusion of this research.

II. RELATED WORK

In recent years, indoor localization has been considered as an important application topic, and much research has been conducted on indoor localization using neural networks. The indoor localization system based on neural networks learns the data features in the training phase to predict the location of the smartphone based on the trained network and the smartphone sensor data in the testing phase [17], [21], [24], [25]. Therefore, we review the research in this field and the existing problems in order to improve the results by providing a more efficient method.

Among the types of work done by recurrent networks in the field of indoor localization using magnetic data. we can refer to [21] written by Zhang et al. In this paper, LSTM networks are used for indoor magnetic localization. Also, the magnetic properties have been analyzed to develop a dimension expansion method based on the double sliding window. Then, to evaluate the performance of the proposed algorithm, they have implemented a magnetic localization method based on LSTMs with fingerprint technique on a real smartphone and compared it with DTW based localization algorithms. In this article, they have used two scenarios with multiple and closed paths to evaluate the performance of their method. Their results show that the LSTM method performs better than the DTW method in terms of the mean and standard deviation of the localization errors, and among the 10 routes in the two scenarios, it has the lowest mean and standard deviation of error of 0.28 m and 0.18 m, respectively.

Reference [25] uses a new TCN multiscale network and LSTM (MSTL) for indoor magnetic localization. They first collected and preprocessed magnetic data from smartphones in experimental environments. Then, they used TCN to expand the dimensions of the data features to evaluate the performance of the proposed algorithm in real environments. This paper has used three experimental localization scenarios with small scale, medium scale and large scale. In this research, using the proposed algorithm, the minimum and maximum standard deviation of localization errors were 0.465 m and 1.111 m, respectively, and the minimum and maximum average localization error in three indoor localization scenarios was 0.75 m and 1.04 m.

In another work [24], deep LSTM is used for indoor localization for two locations, laboratory and corridor. They use data from magnetic and light sensors in smartphones to create two-dimensional data with fingerprinting technique. In the Deep-ML system proposed in this article, it first preprocesses the data and trains the deep LSTM network in the offline phase. Then, they use the newly collected magnetic and light data to estimate the location. They have achieved average distance errors of 0.98 m and 1.36 m in the laboratory and corridor, respectively.

Zhang et al. apply a hybrid algorithm of Wi-Fi features and Pedestrian Dead Reckoning (PDR) features for indoor localization. They have used the KNN fingerprinting method for indoor positioning of smartphones with Wi-Fi networks. They have used a sliding window-based displacement method to generate data features and LSTM network for localization. They have used three scenarios in the test environment, one of which is straight movement in a long path, the other is continuous turning movement in a long path, and the next scenario is localization in both modes [17]. According to the results obtained from the localization scenarios of the direct path, the continuous rotation movement and the rotating movement of the smartphone have achieved an average error of 0.42 m, 1.35 m and 0.51 m, respectively.

The paper [38] explores an indoor magnetic positioning system using CNN and GRU With fingerprint technique, collecting smartwatch and smartphone data in a 185.12 m2 indoor environment. Data were collected from 325 points, spaced 0.6 meters apart, in a location consisting of three corridors, two offices, and a hall. Using only smartwatch magnetic field data, with an 80/20 training/validation split, the authors' CNN (three convolutional layers and four FC layers, each with convolution, activation, max pooling, and elimination stages) achieved an RMS, mean, and standard deviation errors are equal to 0.47 m, 0.383 m, and 0.55 m, respectively. Our work improves upon this by using a larger, more diverse dataset and achieving significantly lower RMSE and STD values, while also considering environments with greater local diversity.

The paper [39] presents an indoor multi-source positioning system that fuses PDR, magnetic fingerprinting, and Wi-Fi fingerprinting data. The PDR module estimates pedestrian position using a walking-based approach. The magnetic

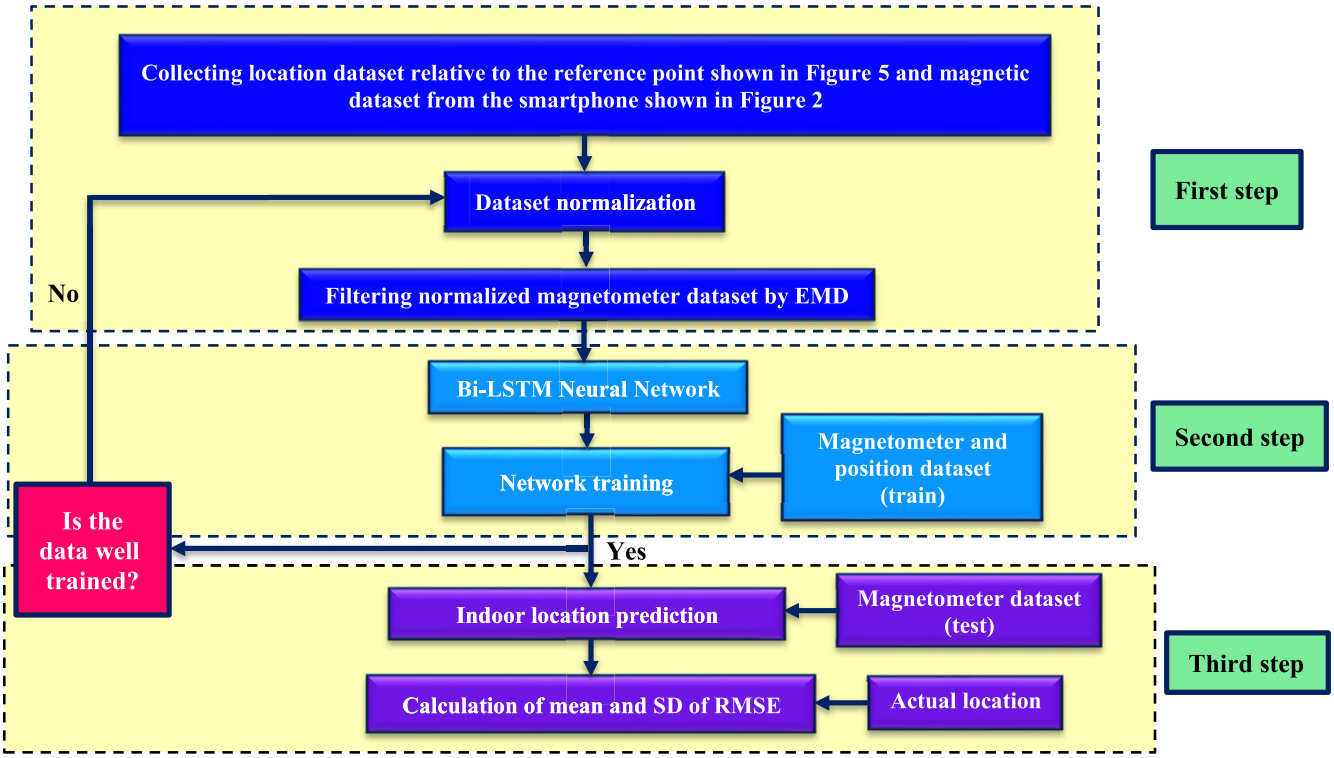

FIGURE 1. Block diagram of the proposed methodology for prediction of the indoor location of the person.

module employs magnetic fingerprints in the offline phase and Elastic Dynamic Time Warping (EDTW) in the online phase for fingerprint matching, followed by KNN for position estimation. Similarly, the Wi-Fi module uses Wi-Fi fingerprint data in the offline phase and Euclidean distance in the online phase for fingerprint matching, also using KNN for position estimation. A robust extended Kalman filter (REKF) fuses the PDR, magnetic, and Wi-Fi data to reduce position error and improve accuracy. Walking experiments were conducted in an educational building, a study room, an office building, and a shopping mall using two pedestrians with smartphones. The proposed model achieved RMSE of 0.81 m, 2.35 m, 1.7 m, and 4.01 m in these locations, respectively, which are significantly different from the results obtained in our research.

In [40], a combination of Wi-Fi, magnetic, and PDR data is used for indoor positioning with fingerprint technique. They use the adaptive extended Kalman filter (AEKF) to combine the information. In general, the positioning area in this experiment is divided into 10 sub-areas. The system is divided into two stages. In the offline stage, the localization area is divided into several sub-areas through RSSI data clustering. Then, the magnetic data is collected, and the RSSI data is trained to build an area discrimination model after extracting the localization sub-area feature. In the offline stage, prediction is performed based on the online RSSI data, and AEKF is used to combine the positioning result. The experiment was conducted on the fourth floor of the School of Environment and Spatial Information, China University of Mining and Technology, which consisted of four corridors and three turns. The positioning error result of the proposed method reached RMSE of 0.81 m, which is a large error compared to our work. In addition, this paper only evaluated one location.

In [41], an indoor positioning method based on an improved particle filter algorithm is used to improve the indoor positioning performance, which combines geomagnetic field data, PDR, and Wi-Fi with fingerprint technique. The authors use Wi-Fi to implement the initial PDR positioning and geomagnetic signals to correct the PDR and Wi-Fi positioning results. They collect Wi-Fi and geomagnetic signals from a smartphone. When the system locates, the acceleration and gyroscope signals are stored in a database to prepare for PDR calculation. Based on the combined positioning approach, the pedestrian position is calculated. The authors were able to achieve an RMSE of 1.91 m with their proposed method, which has a very large error compared to the method proposed in this study.

Considering the reviews that have been done in the field of previous works, the problems in them and their results. We should be able to provide more accurate results in this field using our proposed method. In this study, we use Bi-LSTM neural network to improve the localization performance in indoor environments using smartphone magnetic data because this neural network minimizes the error between the predicted position and the actual position of the user and

increases the localization accuracy and obtains more accurate results than previous works.

III. METHODOLOGY

Figure 1 provides a brief overview of the proposed methodology in this study. Our method is basically based on three steps. In the first step, we collect the location dataset relative to the reference point shown in Figure 5 and the magnetic dataset from the sensor of the Samsung Galaxy A3 smartphone shown in Figure 2. Then we normalize the data and filter the magnetic data using the EMD algorithm (see reference [42; 43] for algorithm details). We design the EMD method for noise removal and processing in magnetic data. In addition, we normalize the data to avoid overfitting in neural networks. After that, we divided the data into two parts: training and testing. In the second step, we use the Bi-LSTM neural network to train the processed magnetic data and the position data. In the last step, the indoor location prediction of the persons using the trained network and magnetometer data is discussed and the mean and standard deviation of RMSE between the actual location and the predicted location are calculated. Our proposed model is indoor localization using Bi-LSTM neural network for smartphone magnetic sensor data using fingerprint method.

A. DATA PREPROCESSING

For data preprocessing, we first collect the location data relative to the reference point and magnetic data from the sensor of the Samsung Galaxy A3 smartphone. Datasets include 1401 data for the localization of indoor space in the engineering faculty of this University. These datasets have been collected from three different places that are: 254, 462 and 685 rows of data from the network and embedded systems research laboratory location and the corridor environment on the second floor of the engineering faculty and a large environment on the first floor of the same faculty for Locating people, respectively. Then normalization is done for data and since the raw data from smartphone sensors are noisy, we filter the magnetic dataset using the EMD algorithm.

In this study, the data is divided into two sets of training and testing, and the validation data is considered as part of the training dataset in order to tune the model and determine the optimal parameters appropriate for the proposed model (learning rate, number of layers, number of neurons, etc.). We examine the output results with optimal parameters to determine the best parameters. In this way, the model performs well on unseen data and prevents overfitting.

B. NETWORK TRAINING

The purpose of this step is to train the data for all three locations used for localization. We use most of the data for training and less than 50 data in each of the three locations for testing. The goal of this work is to use more data for training so that the Bi-LSTM neural network can be trained better and make better prediction. We use Bi-LSTM neural network to learn the features of smartphone magnetic data and location data. Therefore, the network training consists of examining the features of the dataset and examining the Bi-LSTM neural network.

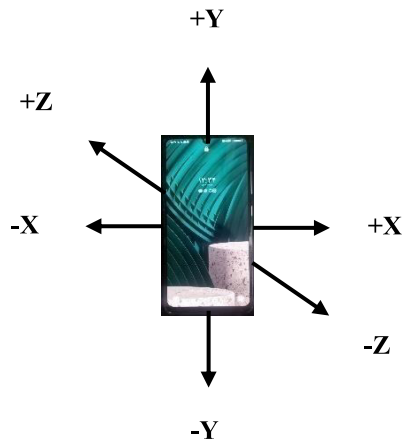

FIGURE 2. Smartphone coordinate system.

One of the problems of neural networks is the overfitting problem, which occurs when the model fits too tightly to the training data and cannot perform well on the test data. Various methods can be used to deal with this problem. One of these methods is to use diverse data and a sufficient number of data to predict internal localization. This problem occurs when the number of data is not enough or does not have enough diversity, because in this case the model may only learn certain patterns and have less accuracy on the test data. However, in this study, we have collected the maximum number of data points in three scenarios at different locations, which increases the number of data and its diversity helps the model identify more realistic patterns. Also, the use of data preprocessing techniques, which include normalization or standardization of data, is effective in improving performance and preventing overfitting. Moreover, in this paper, we have used a Bi-LSTM neural network layer which reduces the probability of overfitting. Therefore, the Bi-LSTM neural network extracts useful features for effective training of the model to make more accurate predictions for the test data.

1) DATASET FEATURES

With the development of smartphones, various sensors have been added to it. Due to their popularity among people, small physical size, light weight and easy portability, these phones can be used as an ideal platform in indoor localization applications. Magnetometer is one of the sensors used in smartphones. This sensor measures all values of the environmental magnetic field in three axes with the unit of μT [24]. The smartphone magnetometer measures magnetic data in vector form in the east, north and vertical directions with three components t_x, t_y and t_z, respectively. Figure 2 shows the different directions of the magnetic data on the smartphone coordinates [24].

We collect location data relative to the reference point simultaneously with the magnetometer data from the smartphone. This means that each magnetometer data is collected simultaneously with a location data. Each location data includes values in the x and y directions that represent a position in the evaluated locations. The location data collected on the first floor of this college also includes the z component because its data is on a lower floor than the reference point.

Most smartphones have built-in accelerometer, gyroscope, and magnetometer sensors that are used for various applications such as positioning and tracking. These sensors measure acceleration, angular velocity, and magnetic field, respectively. The advantage of using smartphone sensors is that they do not require additional hardware to be installed for indoor positioning. In addition, they are lightweight, inexpensive, and readily available.

We can use various methods to recover data from our smartphone. Here, we will explore methods based on information obtained from built-in sensors, user input, and connected wearable devices:

1. Data retrieval via internal motion sensors (accelerometer, gyroscope, etc.):

Motion pattern retrieval: Motion sensors can record the user's motion patterns. These patterns can be used to identify daily activities. If the user has lost information, the recorded motion patterns can be used as clues to reconstruct the user's activities at a specific time. For example, if the user cannot remember where they saved a file, examining the motion patterns can reveal whether the user was walking around the house or sitting at work at that time.

Approximate location retrieval: With magnetometer, accelerometer, and gyroscope data, the user's approximate location in a limited environment can be estimated. This method can be used to track the user's movement path in a building or indoor environment. This information can be used to find the exact location of a lost photo or file.

2. Recover data through user input:

Search history: Checking the search history in apps and browsers can help find information that the user has previously searched for. If the user has forgotten the name of a file or website, checking the search history can help find it.

Checking data stored in apps: Many apps store information about the user's activities. Checking this information can help find lost data.

If the user has created a file in a specific app, checking the data stored in that app can help find the file.

Using backup data: If the user has backed up their phone data, they can use this backup to recover lost data.

This is one of the most reliable methods of data recovery, provided that the user has regularly backed up their data.

3. Recovering data through connected wearable devices:

Data synchronization:

Many of the user's wearable devices (such as smartwatches) sync data with their smartphone. This data can include information about exercise activities, sleep, heart rate, etc. If the user has lost their phone data, they may be able to recover it via their wearable device.

2) BI-LSTM NEURAL NETWORK

We first explain the structure of the LSTM neural network so that we can use it to describe the structure of the Bi-LSTM neural network. LSTM neural network is a special type of RNN networks. The main idea of LSTM neural network is a gated logic. this network enables a memory-based architecture and because of having a specific memory cell, they can solve the problem of long-term dependencies [21]. Figure 3 shows the LSTM neural network. First, the LSTM neural network decides what information to remove from the cell state. This decision is made by the forgetting gate layer. The hidden state information of the previous step (H_{t-1}) and the input information of the new step (M_t) are the inputs of the forgetting gate layer, perform the process of removing or keeping information from the cell state. The output of F_t is a number between 0 and 1. This process is shown below:

$$F_t = \sigma\left(W_F.[H_{t-1}; M_t] + B_f\right) \tag{1}$$

In the next step, the LSTM neural network decides what new information to store in the cell state. this function decides what information to update (I_t), Then, create a new vector of candidate values C_t^*. Finally, the sigmoid function decides which values of the tanh function output should be kept. This process is shown below:

$$I_t = \sigma\left(W_I.[H_{t-1}; M_t] + B_i\right) \tag{2}$$

$$C_t^* = \tanh(W_c.[H_{t-1}; M_t] + B_c) \tag{3}$$

And then, update C_t. The process is written as follows:

$$C_t = F_t * C_{t-1} + I_t * C_t^* \tag{4}$$

Finally, the new C_t value and the new hidden state (H_t) are determined. Sigmoid function is executed to decide which parts of C_t are sent to the output, the output of this function is called O_t. Then to determine the new H_t value. The process is written as follows:

$$O_t = \sigma\left(W_0.[H_{t-1}; M_t] + B_0\right) \tag{5}$$

$$G_t = \tanh\left(C_t\right) \tag{6}$$

$$H_t = O_t * G_t \tag{7}$$

where, W_F, W_I, W_C and W_O represent the weight matrices of the forget gate, input gate, memory cell gate and output gate, respectively. Also, B_f, B_i, B_c and B_0 represent the biases for the forget gate, input gate, memory cell gate and output gate, respectively. σ and tanh represent the sigmoid and hyper tangent nonlinear activation functions.

The above steps introduce the basic structure of LSTM neural network. Because this network can store information for a long time, it is used for long term forecast. As a result, it solves the problem of long-term data storage.

Bidirectional RNN (Bi-RNN) are a type of RNN networks and have two hidden layers in different directions in the neural networks and enter the two hidden layers to the same

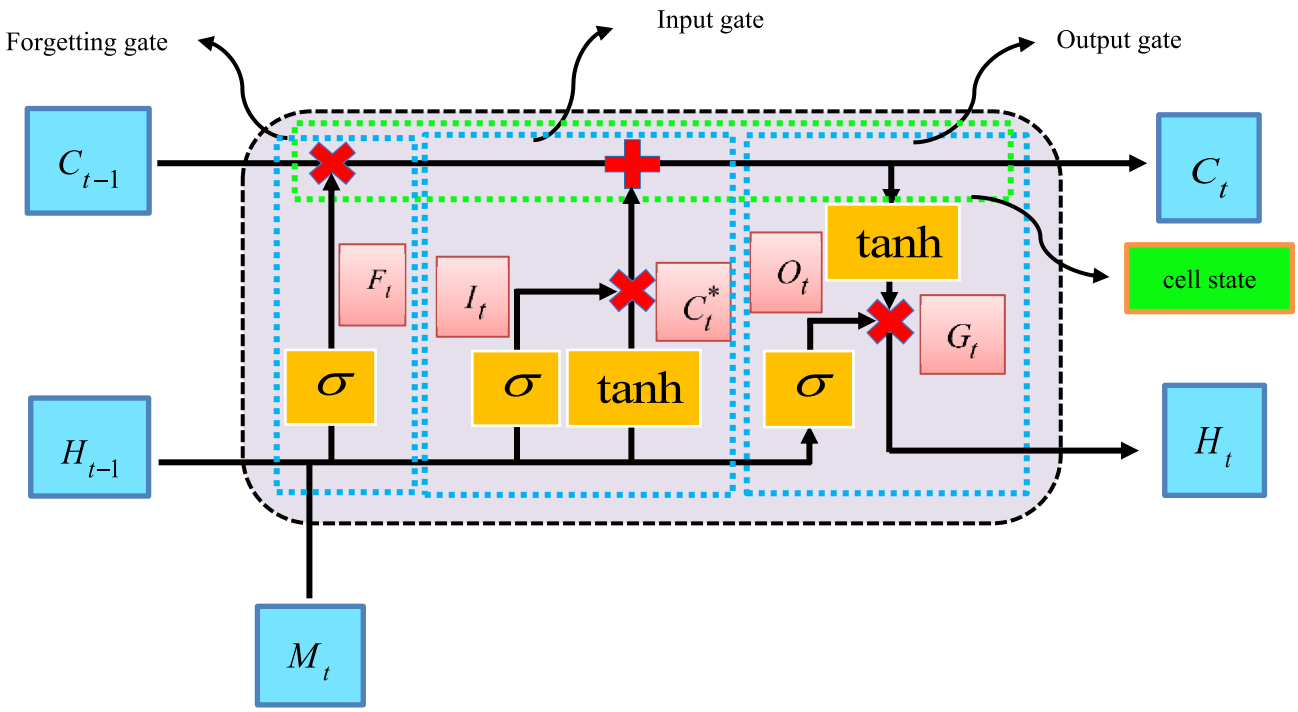

FIGURE 3. Structure of LSTM neural network.

output layer [33], [34], [35], [36], [37]. However, other neural networks such as FFN [44], [45] have the limitation of the stability of the input data. In addition, this neural network does not have the ability to receive information from the future, and its other weakness is not having a memory to store information. on the other hand, RNN networks such as: LSTM and GRU [46] neural network also have limitations because they cannot receive data from the future.

To eliminate these problems, Bi-RNN networks have been proposed to overcome the aforementioned limitations. Bi-LSTM is a Bi-RNN network and unlike regular LSTM which has one LSTM layer, it has another LSTM layer. It should be noted that each layer has its own hidden states and memory cells. It should be noted that each layer has its own hidden states and memory cells. The structure of Bi-LSTM neural network is shown in Figure 4.

C. NETWORK TEST

In the network testing phase, we take less than 50 data points from the processed magnetic dataset for testing. We use these data points together with the trained network to predict the future position. Also, to evaluate the effectiveness of the proposed method, we use the RMSE mean and SD method to determine the error between the actual position (P_A) and the predicted position (P_P), which we calculate as follows:

$$RMSE(P_A, P_P) = \sqrt{\frac{1}{c}\sum_{k=1}^{c} \|P_A - P_P\|^2} \tag{8}$$

IV. EXPERIMENT RESULT AND DISCUSSION

A. EXPERIMENTAL RESULTS

In this section, LSTM, Bi-LSTM, GRU and FFN neural networks are used for indoor localization. First, we conduct several experiments to investigate the effect of each parameter on the test results to compare the performance of named neural networks for more accurate indoor localization in three different locations. We have collected dataset of smartphone magnetic sensor from three different locations in Engineering Faculty of this University, as shown in Figure 5, We have used from a small location in the Network and Embedded Systems Research Laboratory on the second floor of this Faculty with several spiral paths (Figure 5a), a large closed place in the corridor of the same floor (Figure 5b) and a large place on the first floor of the same Faculty (Figure 5c) to evaluate different methods. All programs were implemented using MATLAB software on windows 10 64-bit system. It has a 10th generation Intel (R) Core (TM) i3 processor with a base frequency of 1.20 GHz and 20 GB of RAM. Run time for all programs took only a few seconds.

We consider the distance between each step and the next step at all three study locations to be 0.3 meter and use a reference in all three experiments to fairly evaluate the performance of all three locations. The first experiment is related to the network and embedded systems research laboratory with area of 8.2 × 5.7 square meter on the second floor of the Engineering Faculty of this University. We select 220 training data and 34 test data from the 254 data collected from this location. The second experiment is related to the corridor on

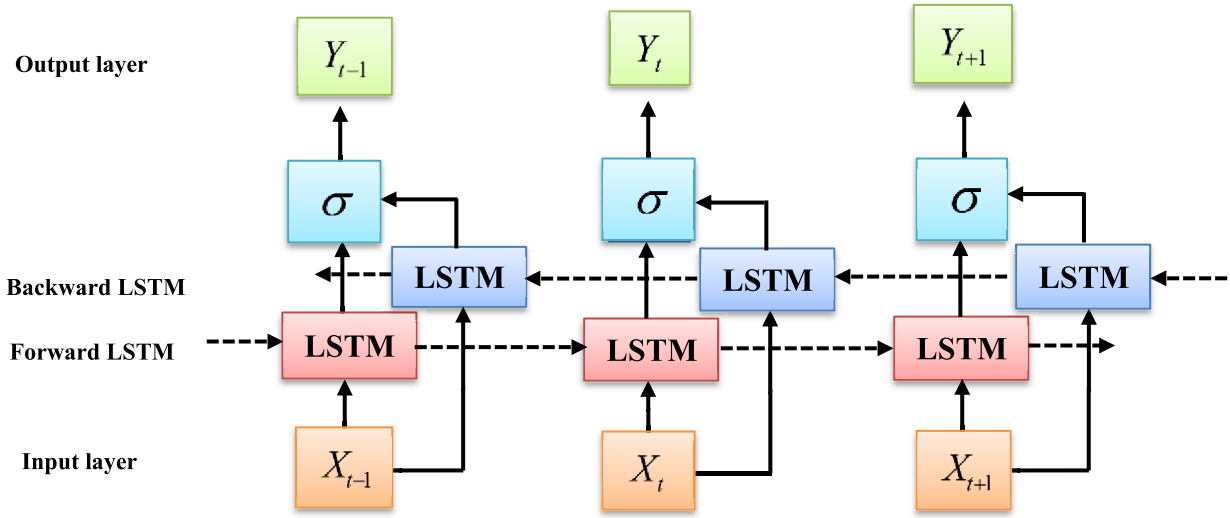

FIGURE 4. Structure of BI-LSTM neural network.

the same floor. It is a long and empty corridor. We select 439 training data and 23 test data from the 462 data collected from this location. The third experiment is related to the first floor in the same faculty. This location has a large area and we have tried to collect data from all point of this location. We select 650 training data and 35 test data from the 685 data collected from this location.

We use the device information of the smartphone to collect magnetometer data from its magnetic sensor, and the data is collected by entering the page containing the number of sensors and applications or by entering the code ∗#0∗# and entering the sensors section.

The indoor localization performance is affected by the length of magnetic data and the internal architecture of neural networks. The number of hidden units (neurons) and length of magnetic data are two key factors to determine the indoor localization in these networks. This article analyzes the two key factors for better performance. The number of hidden units is an important factor that affects the localization accuracy. In this examination, the hidden unit range is set from 20 to 400 and test data size is set from 4 to 84. Hyper-parameter tuning for network design was done based on the final results and we considered the settings to be appropriate for the best results. Figure 6 shows the mean RMSE of indoor localization using LSTM, Bi-LSTM, GRU and FFN neural networks for different number of hidden units and different sizes of test data in the laboratory location.

As can be seen in Figure 6a, the mean RMSE of indoor localization using RNN networks for laboratory location in all hidden units is less than 0.3 m. Therefore, this type of neural network has a very good performance compared to other neural networks. Also, the minimum mean RMSE of indoor localization for LSTM, Bi-LSTM, GRU and FFN neural networks is 0.15 m with 220 hidden units, 0.12 m with 30 hidden units, 0.19 m with 270 hidden units and 0.18 m with 5 hidden units, respectively. Therefore, the Bi-LSTM neural network has less error than other RNN networks. So, according to the results obtained from Figure 6a for different hidden units, the Bi-LSTM neural network has been able to successfully predict the position of the person in navigation mode. The neural network's limited number of hidden units leads to a quick response time, enhancing its performance.

As can be seen in Figure 6a, the mean RMSE of the indoor localization of the Bi-LSTM neural network for all hidden units is lower than that of the other neural networks used in this study. The mean RMSE of the Bi-LSTM network decreases up to 200 hidden units and then increases. This means that this neural network can improve indoor localization accuracy by increasing the number of hidden units in a certain range.

Also, due to the limited data available in this location.; the GRU neural network performed well for different numbers of hidden units and had the minimum mean RMSE at 400 hidden units, which is due to the large number of hidden units, which causes the response time for this neural network to be long. The LSTM neural network, like the GRU neural network, also has the minimum mean RMSE at 400 hidden units, which causes the response time to be long. For the FFN neural network, the indoor localization error increases with the increase in the number of hidden units and has the highest mean RMSE with 300 hidden units. Therefore, the Bi-LSTM neural network outperformed all the neural networks in terms of both response time and mean RMSE.

Figure 6b illustrates the mean RMSE of LSTM, Bi-LSTM, GRU, and FFN neural networks with test data sizes of 4, 14, 24, 34, 44, 54, 64, 74, and 84 for the laboratory location. The mean RMSE of the Bi-LSTM neural network for this location decreases with test data size up to 24 but increases for larger sizes. Figure 6b shows that Bi-LSTM has a lower mean RMSE than other neural networks across all test data at

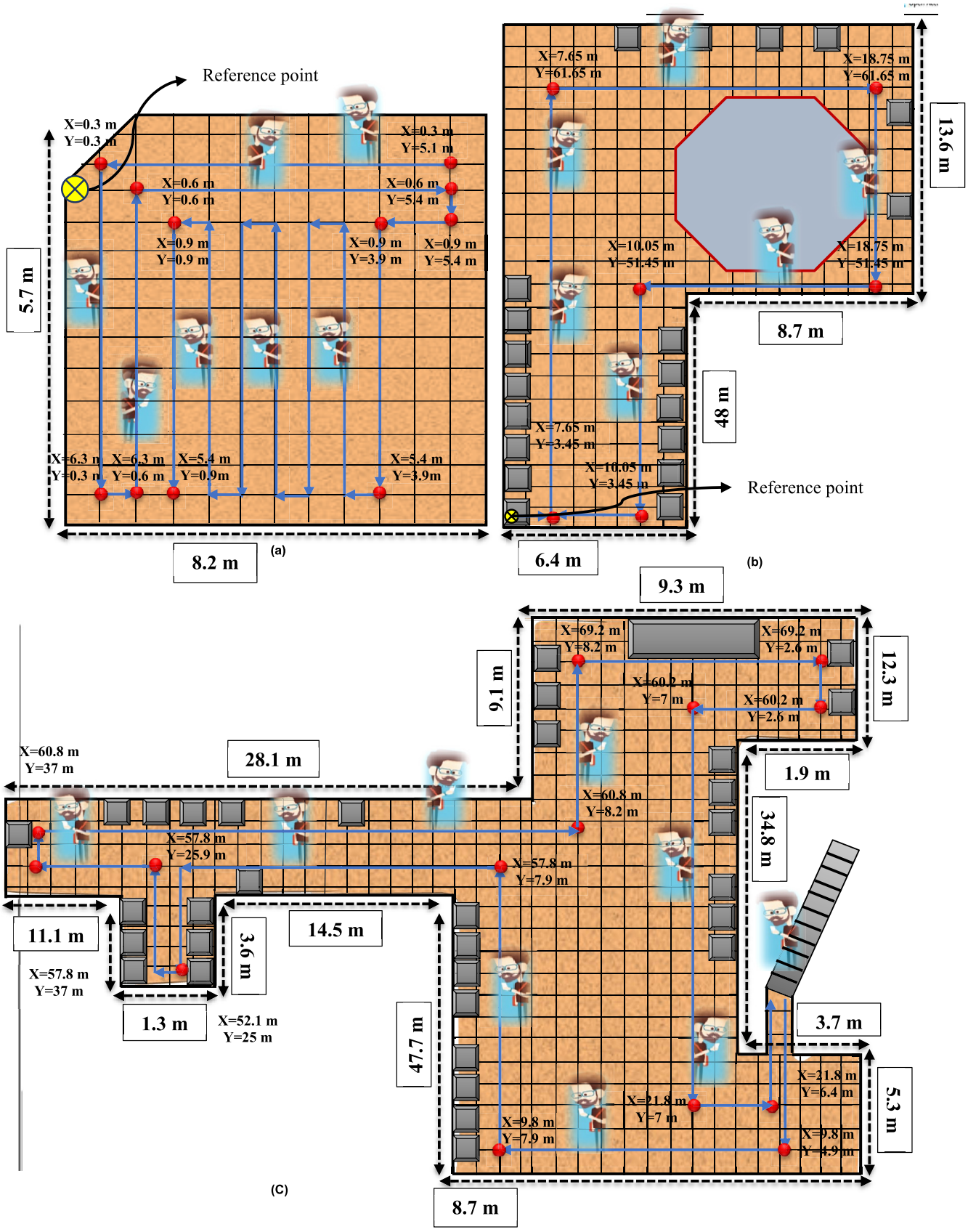

FIGURE 5. Data collection locations: a) Network and embedded systems research laboratory location, b) Corridor location, and c) First floor location.

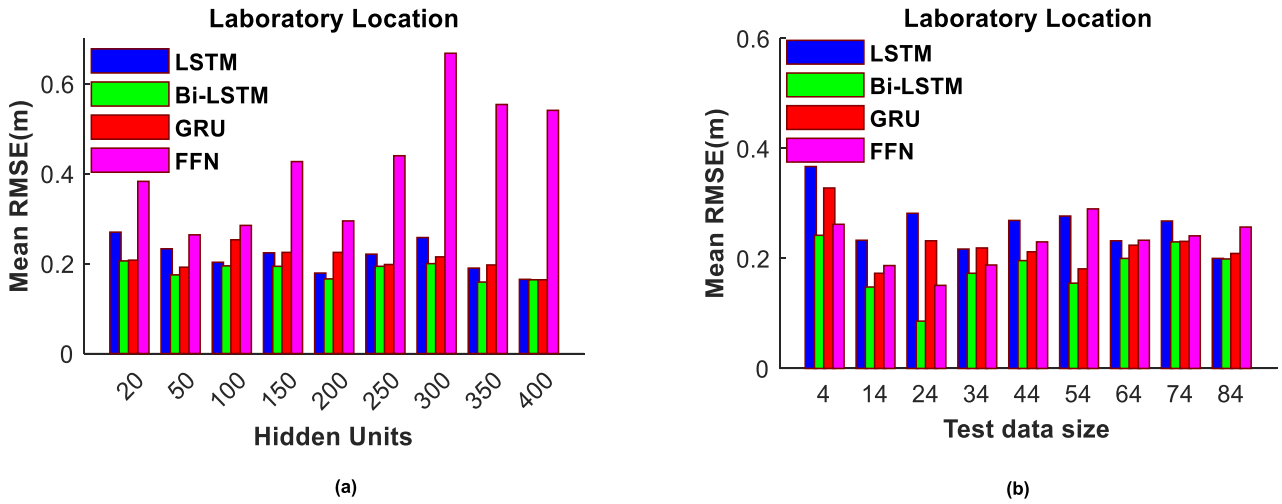

FIGURE 6. Mean RMSE of indoor localization using LSTM, Bi-LSTM, GRU and FFN neural networks in the location laboratory for: a) different number of hidden units and b) different size of test data.

this location. The FFN network has the lowest mean RMSE at a test data size of 24 in the laboratory location, and the indoor localization error increases with increasing test data size up to 54. The GRU and LSTM networks achieve the lowest mean RMSE for test data sizes of 14 and 84, respectively.

Figure 6 shows that the Bi-LSTM neural network outperforms other neural networks for different hidden units and different test data. The minimum and maximum mean RMSE of indoor localization for this neural network are 0.12 m and 0.24 m, respectively. These results show that the indoor localization performance of the Bi-LSTM neural network is very strong. In addition, the proposed method can achieve higher accuracy for larger training data. The computation time for each test data size is only a few seconds. Therefore, this is a very fast response time for indoor localization. Figure 7 shows the mean RMSE of indoor localization using LSTM, Bi-LSTM, GRU, and FFN neural networks for different numbers of hidden units and different test data sizes in the corridor location.

Figure 7a shows that the mean RMSE of indoor localization using RNN networks for corridor location in all hidden units is less than 0.18 m. Therefore, this type of neural network has performed very well compared to other neural networks for this location as well as the laboratory location. The mean RMSE of LSTM, Bi-LSTM and GRU networks decreases up to 50, 150 and 50 hidden units, respectively and then increases slightly and remains almost constant. This means that these neural networks can improve the localization accuracy by increasing the number of hidden units in a certain range. Considering that more data is used in this place than in the laboratory place; the results of the RNN neural networks in this place are significantly improved compared to the laboratory location. so, this type of neural network has been able to achieve a lower mean RMSE in this location than in the laboratory.

So, in the location minimum mean RMSE of indoor localization for LSTM, Bi-LSTM, GRU and FFN neural networks is 0.05 m with 50 hidden units, 0.02 m with 150 hidden units, 0.05 m with 70 hidden units and 0.68 m with 6 hidden units, respectively. The Bi-LSTM neural network has lower error than other RNN networks and has a mean RMSE of indoor localization less than or equal to 0.1 m for number of different hidden units. Therefore, according to the results obtained for different hidden units, the Bi-LSTM neural network has been able to successfully predict the position of the person in navigation mode. Also, the FFN neural network did not perform well for all the hidden units and achieved the minimum and maximum mean RMSE of 0.67 m with 200 hidden units and 1.45 m with 50 hidden units, respectively.

Figure 7b illustrates the mean RMSE of LSTM, Bi-LSTM, GRU, and FFN neural networks with test data sizes of 4, 14, 24, 34, 44, 54, 64, 74, and 84 for the corridor location. According to the results, the mean RMSE of RNN neural networks increases with the increase of the test data size. The Bi-LSTM neural network reduces the indoor localization error by increasing the size of the test data up to 24; Then, the positioning error tends to increase for test data greater than this value. Therefore, we have a lower mean RMSE of indoor localization in RNN neural networks for less test data. The FFN neural network has the lowest mean RMSE for test data 4 and has performed poorly compared to the RNN neural network, and the error amount has increased significantly with this method. As you can see, the Bi-LSTM neural network has a lower mean RMSE than the other neural networks examined in this study for all test data in the corridor location and has achieved very good results.

Based on the results obtained from these four methods for two different locations, the performance of the Bi-LSTM neural network has been better than other methods for indoor localization. However, GRU neural network is more useful

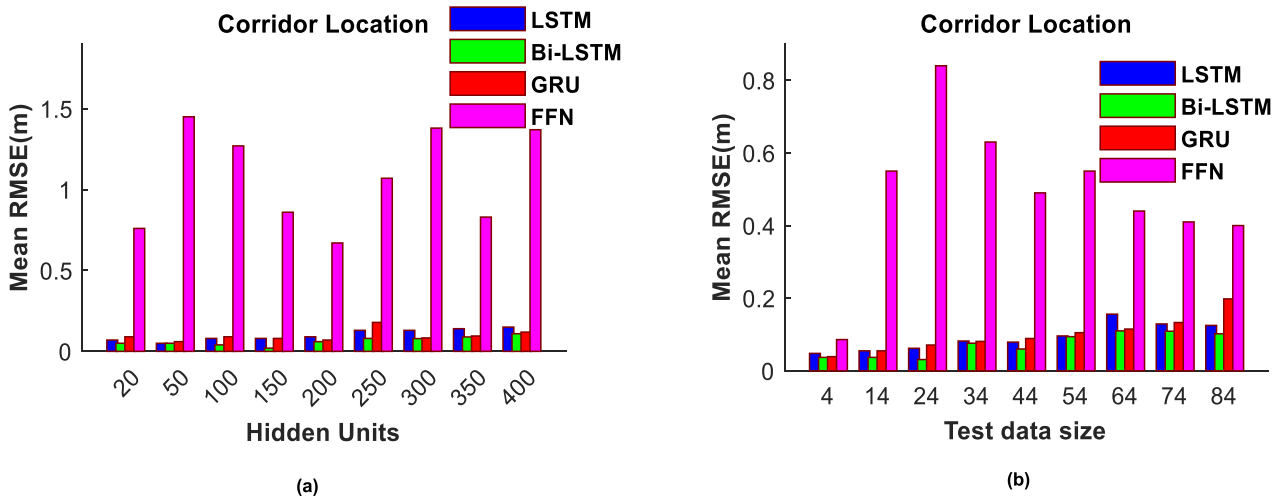

FIGURE 7. Mean RMSE of indoor localization using LSTM, Bi-LSTM, GRU and FFN neural networks in the location corridor for: a) different number of hidden units and b) different size of test data.

when less training data is available due to the smaller number of parameters. But LSTM neural network is preferred for larger number of training data. The FFN network is a neural network that does not create any loop between its different parts. Therefore, information only moves in the forward direction. For this reason, this neural network does not perform well compared to RNN networks. Figure 8 shows the mean RMSE of indoor localization using LSTM, Bi-LSTM, GRU and FFN neural networks for different number of hidden units and different sizes of test data in first floor location.

Figure 8a indicates that Bi-LSTM neural network performs better for all hidden units than other neural networks used in this article and has achieved a lower mean RMSE. Also, RNN neural networks have less error and better indoor localization than FFN neural network for all hidden units. The minimum mean RMSE of indoor localization for LSTM, Bi-LSTM, GRU and FFN neural networks is 0.04 m with 10 hidden units, 0.01 m with 130 hidden units, 0.05 m with 100 hidden units and 0.18 m with 7 hidden units, respectively.

Therefore, according to the results obtained for different hidden units, the Bi-LSTM neural network has been able to successfully predict the position of the person. Also, this neural network achieved the minimum and maximum mean RMSE of 0.02 m with 20 hidden units and 0.09 m with 150 hidden units, respectively. As you can see in this figure, the RNN neural network performed better for all the hidden units than the FFN neural network. FFN neural network did not perform well for all the hidden units and achieved the minimum and maximum mean RMSE of 0.19 m with 20 hidden units and 0.34 m with 400 hidden units, respectively.

Therefore, according to the results obtained for different hidden units, the Bi-LSTM neural network has been able to successfully predict the position of the person. Also, this neural network achieved the minimum and maximum mean RMSE of 0.02 m with 20 hidden units and 0.09 m with 150 hidden units, respectively. As you can see in this figure, the RNN neural network performed better for all the hidden units than the FFN neural network. FFN neural network did not perform well for all the hidden units and achieved the minimum and maximum mean RMSE of 0.19 m with 20 hidden units and 0.34 m with 400 hidden units, respectively.

The LSTM neural network has the minimum error for 20 hidden units and its error increases with increasing number of hidden units. But the GRU neural network has the minimum error for 300 hidden units and its error decreases with increasing number of hidden units. The error of Bi-LSTM neural network increases up to 150 hidden units and for more hidden units the error decreases and then remains almost constant.

Figure 8b illustrates the mean RMSE of LSTM, Bi-LSTM, GRU, and FFN neural networks with test data sizes of 4, 14, 24, 34, 44, 54, 64, 74, and 84 for first floor location. The minimum mean RMSE of indoor localization for LSTM, Bi-LSTM, GRU and FFN neural networks is 0.03 m with 4 test data, 0.02 m with 64 test data, 0.05 m with 14 test data and 0.16 m with 84 test data. In addition, Bi-LSTM neural network performs better than the other neural networks on all test data and has lower indoor localization error. Therefore, according to the results obtained both for different hidden units and for different test data sizes, this neural network provides more correct results than other neural networks.

According to the results obtained and the investigations done from three different locations for indoor localization using LSTM, Bi-LSTM, GRU and FFN neural networks, the Bi-LSTM neural network has higher accuracy and lower mean RMSE than other neural networks both for different hidden units and for different test data. Therefore, the proposed method can be used for real indoor localization applications.

Table 1 draws a quantitative comparison between the neural networks mentioned above based on the RMSE metric.

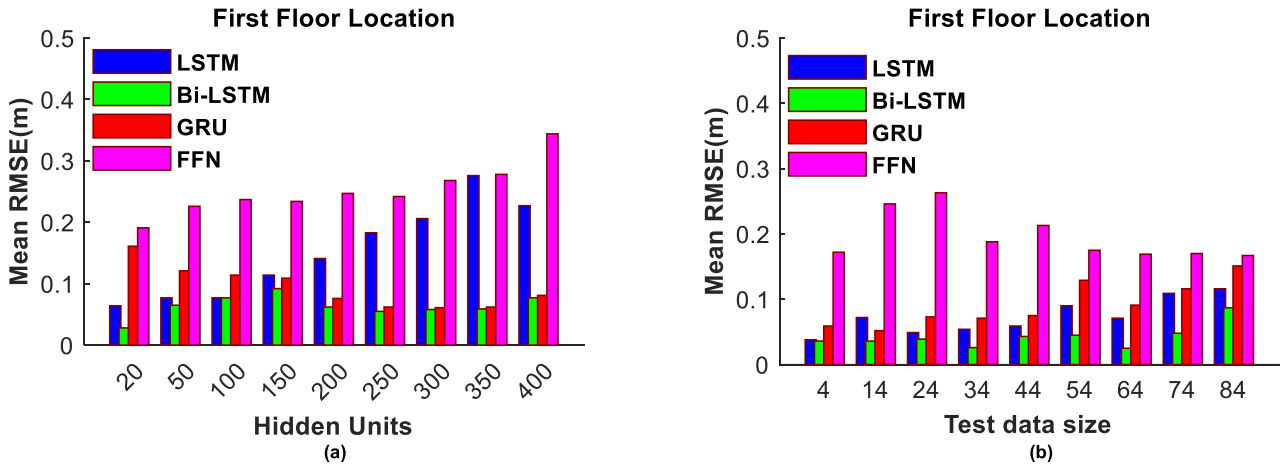

FIGURE 8. Mean RMSE of indoor localization using LSTM, Bi-LSTM, GRU and FFN neural networks in the location first floor for: a) different number of hidden units and b) different size of test data.

TABLE 1. A comparison of the mean and SD of the RMSE error between LSTM, Bi-LSTM, GRU and FFN neural networks in laboratory, corridor and first floor locations.

Location	Algorithm	RMSE (m)	
		Mean	SD
Laboratory	LSTM	0.1547 m	0.1402 m
	Bi-LSTM	**0.1244 m**	**0.1070 m**
	GRU	0.1939 m	0.1135 m
	FFN	0.1855 m	0.1320 m
Corridor	LSTM	0.0564 m	0.0372 m
	Bi-LSTM	**0.0251 m**	**0.0166 m**
	GRU	0.0549 m	0.0340 m
	FFN	0.6809 m	0.1942 m
First floor	LSTM	0.0454 m	0.0228 m
	Bi-LSTM	**0.0192 m**	**0.0165 m**
	GRU	0.0585 m	0.0229 m
	FFN	0.1832 m	0.1230 m

The Results show that the Bi-LSTM neural network in comparison with LSTM neural network reduces the RMSE errors by 19.58%, 55.49% and 57.70% for laboratory, corridor and first floor locations, respectively; Bi-LSTM neural network in comparison with GRU neural network reduces RMSE errors by 35.84%, 54.28% and 67.17% for laboratory, corridor and first floor locations, respectively; and the Bi-LSTM neural network in comparison with FFN neural network lowers the RMSE errors by 32.93%, 96.31% and 89.51% for laboratory, corridor and first floor locations, respectively. As the results show, the Bi-LSTM neural network has significantly improved over the other neural networks examined in this study for all three locations.

The laboratory location data is collected from complex paths in a small space. In contrast, the corridor location and first floor location data are collected from a much larger area than the laboratory location, which has more data than this location. Since neural networks are good performance of fingerprinting techniques in localization systems [6, 8, 16-17, 21, 24, 38-39, 40-41, better trained with more data and have higher prediction accuracy, the indoor location prediction results for the corridor location and first floor location are better than the laboratory results in terms of mean and SD of RMSE.

B. DISCUSSION

In this study, our method is to predict the location of people based on Bi-LSTM neural network using real magnetic dataset and Fingerprint technique, that is, Bi-LSTM neural network model of magnetic fingerprint based on meter parameter. Our proposed method meets the requirements of indoor localization. The location prediction results show that the proposed model can predict the location of people well. Table 2 Comparison between our proposed work and previous papers based on location techniques, location algorithm, data type, number of locations and results is given.

As can be seen in table 2, previous studies of indoor positioning based on smartphone data [21], [24], [38], [40], [41], [47] have often used magnetometer data alone or in combination with other data and have employed fingerprinting techniques. This is due to the accuracy and [47]. In this study, we have also used magnetometer data and fingerprinting techniques. Furthermore, while previous studies have evaluated their models at one or two locations, we have tested the model at three different locations to ensure its correct performance.

Our goal in this paper is to use an algorithm for indoor localization that has the most accurate results and the least error. In the previous works listed in the table above, most of them have used neural networks as their proposed algorithm. In this paper, we use an algorithm based on Bi-LSTM neural network and fingerprinting technique with magnetometer data for indoor localization. This approach is inspired by reference [21], which used LSTM neural network and magnetometer data to achieve more accurate results and less error. The research results show that our proposed method significantly reduces the indoor localization error compared to other methods and is a step forward in this field.

TABLE 2. The proposed method and other works comparison.

Authors	Year	Localization algorithm	Positioning technique	data type	Number of places	Results
Wang et. al. [24]	2020	Deep LSTM network	Fingerprint	Magnetic and light data in smartphones	2	The average errors at the laboratory and corridor locations are 0.98 m and 1.36 m, respectively.
Zhang et al [21]	2021	LSTM networks	Fingerprint	Smartphone magnetic data	1	The minimum mean and standard deviation of error are 0.28 m and 0.18 m, respectively.
Abid et al [38]	2021	CNN-GRU	Fingerprint	Smartwatch magnetic data	1	The RMS, mean, and standard deviation errors are equal to 0.47 m, 0.383 m, and 0.55 m, respectively.
Wang et al [40]	2022	Wi-Fi\MMS\PDR+ AEKF	Fingerprint	PDR, magnetic, and Wi-Fi dataset	1	RMSE of 0.81 m
Wang et al [41]	2024	Improved particle filter	Fingerprint	PDR, magnetic, and Wi-Fi dataset	1	RMSE of 1.91 m
Zhou et al [44]	2024	Two-layer LSTM network and CNN	Factor graph optimization	Accelerometer, gyroscope, magnetometer, and gravity sensor data	1	The RMSE of the two paths in the four cases of holding the smartphone is more than 1 m.
This work	**2025**	**Bi-LSTM network**	**Fingerprint**	**Smartphone magnetic dataset**	**3**	**The mean and standard deviation of RMSE in the laboratory, corridor, and first floor locations are 0.12 m and 0.10 m, 0.02 m and 0.01 m, and 0.01 m and 0.01 m, respectively.**

To our knowledge, the most important research works on prediction of the persons indoor position are carried out by Zhang et al. [21] and Abid et al. [38]. Zhang et al. predict the location of a person inside a building using fingerprint technique in three steps. They first obtain magnetic data from the magnetometer sensor of a smartphone. Then, they preprocess this dataset, which includes data normalization and development of a dual sliding window-based dimensionality expansion method. In the next step, they feed it as input to LSTM networks to predict the location of the person and implement it on a real smartphone. Finally, they compare the results of their proposed method with DTW-based localization algorithms. They use RMSE to quantitatively evaluate the performance of the proposed method in two scenarios. Their results show that the LSTM method outperforms the DTW method in terms of mean and standard deviation of localization errors, and has the lowest mean and standard deviation of errors among 10 paths in two scenarios with multiple and closed paths, respectively, of 0.28 m and 0.18 m.

Abid et al. used magnetic data from a smartwatch to investigate indoor positioning in a 185.12 square meter indoor environment. Data were collected from 325 points spaced 0.6 m apart in a location consisting of three corridors, two offices, and a hall. They performed prediction using a combination of CNN and GRU with fingerprinting techniques, achieving RMS, mean, and standard deviation errors of 0.47 m, 0.383 m, and 0.55 m, respectively. Our work improves on this by using a larger and more diverse dataset and achieving significantly lower RMSE and STD values, while considering environments with greater local variability.

V. CONCLUSION

Our method in this paper is to predict the indoor location of people using smartphone magnetic data and Bi-LSTM neural network with fingerprint technique. According to the results obtained in this paper, the Bi-LSTM method has shown satisfactory performance for locating people indoors with smartphone magnetic data. The proposed model is able to be trained with the magnetic data collected from the smartphone and the position data and predict the future position. The results show that the Bi-LSTM neural network can predict the indoor position well compared to the GRU, LSTM and FFN neural networks in three different locations in the Faculty of Engineering of this university. Also, compared to the GRU, LSTM and FFN neural networks, the Bi-LSTM neural network provides the most accurate prediction performance, the mean and standard deviation of RMSE for the Network Research Laboratory and Embedded Systems are 0.1244 m and 0.1070 m, respectively, for the corridor location are 0.0251 m and 0.0166 m, respectively, and for first floor location are 0.0192 m and 0.0165 m, respectively. Two locations, the laboratory and the corridor, are located on the second floor of the faculty, and the third location is on the first floor of the faculty. Our findings can be used in practical applications for predicting the internal position of people with

diseases such as amnesia and mental disorders, and predicting the position of people in crowded and security environments.

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